

MAGDEBURG, Germany

WMO: 103610

Lat: 52.1028N

Lon: 11.5828E

Elev: 85

StdP: 100.31

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-12.8	-10.0	-15.8	0.9	-11.1	-12.9	1.2	-9.4	12.4	6.9	10.9	5.0	1.7	60	0.550

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	29.7	19.2	27.8	18.5	26.1	17.6	20.3	27.3	19.3	25.8	18.4	24.5	2.9	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.9	13.0	22.6	16.9	12.1	21.8	15.9	11.4	21.2	58.5	27.1	55.2	25.7	52.3	24.4	24.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.7	8.3	7.2	DB	-13.6	33.5	6.0	1.9	-17.9	34.8	-21.4	35.9	-24.7	37.0	-29.1	38.3	(4)
(5)			WB	-14.0	22.5	5.8	1.0	-18.2	23.2	-21.6	23.7	-24.9	24.3	-29.1	25.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.2	0.6	0.3	4.7	8.1	13.2	15.9	18.1	18.0	14.3	9.9	4.7	2.2	(6)
(7)		DBStd	7.46	5.58	5.21	3.82	3.63	3.60	3.44	3.18	3.09	2.75	3.66	3.69	3.99	(7)
(8)		HDD10.0	1272	292	272	166	78	12	1	0	0	1	47	162	243	(8)
(9)		HDD18.3	3417	549	505	421	306	161	89	44	45	125	262	409	501	(9)
(10)		CDD10.0	992	0	1	3	22	112	178	252	248	129	43	3	1	(10)
(11)		CDD18.3	96	0	0	0	0	3	17	38	34	3	1	0	0	(11)
(12)		CDH23.3	1134	0	0	0	1	68	198	442	371	49	6	0	0	(12)
(13)		CDH26.7	297	0	0	0	0	4	46	131	110	6	1	0	0	(13)
(14)	Wind	WSAvg	3.2	3.8	3.3	3.5	3.2	2.8	2.8	2.8	2.6	3.0	3.1	3.5	3.6	(14)
(15)	Precipitation	PrecAvg	528	39	28	33	28	58	52	64	58	47	38	41	41	(15)
(16)		PrecMax	752	70	54	80	75	159	110	102	117	130	105	90	81	(16)
(17)		PrecMin	395	9	7	4	3	13	19	13	18	11	10	1	13	(17)
(18)		PrecStd	88	18	14	20	16	36	22	27	26	27	24	21	19	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.8	14.9	18.1	22.8	27.1	31.0	31.7	32.7	27.5	23.1	15.0	12.5	(19)
(20)			MCWB	8.8	10.8	11.3	13.6	16.7	19.7	19.8	20.4	18.2	16.8	10.8	9.5	(20)
(21)		2%	DB	9.6	11.3	14.3	19.7	25.4	27.9	29.8	29.6	24.6	19.3	12.7	10.8	(21)
(22)			MCWB	7.7	8.5	9.7	12.3	15.7	18.5	19.4	19.0	17.0	14.5	10.2	8.6	(22)
(23)		5%	DB	8.4	8.6	12.0	17.4	23.3	25.6	28.0	27.2	21.9	17.1	11.2	9.5	(23)
(24)			MCWB	6.5	6.7	8.3	11.1	15.0	17.3	18.9	18.3	15.9	13.2	9.2	7.7	(24)
(25)		10%	DB	7.1	6.6	10.3	14.9	21.2	23.3	25.9	25.2	19.6	15.5	10.0	8.1	(25)
(26)			MCWB	5.3	4.7	7.4	9.7	14.2	16.4	18.0	17.4	14.8	12.3	8.3	6.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.2	11.1	11.7	14.4	18.1	21.4	22.0	21.4	19.2	17.2	11.6	10.0	(27)
(28)			MCDWB	10.8	14.3	16.1	21.3	24.5	27.8	29.8	31.1	25.9	22.0	13.7	12.1	(28)
(29)		2%	WB	7.9	8.5	10.1	12.9	16.8	19.6	20.4	19.9	17.7	15.3	10.4	8.8	(29)
(30)			MCDWB	9.7	10.8	14.1	18.6	23.7	25.9	27.2	26.7	22.9	18.8	12.3	10.7	(30)
(31)		5%	WB	6.7	6.7	8.8	11.5	15.5	18.3	19.4	19.0	16.5	13.7	9.5	7.8	(31)
(32)			MCDWB	8.2	8.6	11.5	16.3	22.0	23.9	26.1	25.4	20.8	16.4	11.2	9.4	(32)
(33)		10%	WB	5.4	4.8	7.6	10.3	14.4	17.0	18.3	18.1	15.5	12.6	8.3	6.6	(33)
(34)			MCDWB	7.0	6.4	10.1	14.6	20.5	22.2	24.6	23.9	19.2	15.0	9.9	8.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.1	6.5	7.5	9.5	10.8	10.1	11.0	11.1	9.5	8.3	5.8	4.6	(35)
(36)			MCDBR	5.5	8.7	10.7	13.4	14.8	14.4	15.5	15.2	13.4	11.1	7.3	5.9	(36)
(37)			MCWBR	4.7	6.5	7.0	7.2	7.3	6.8	6.9	6.8	7.2	6.9	5.8	4.7	(37)
(38)		5% WB	MCDWB	5.7	8.7	10.3	11.9	13.5	12.8	13.7	12.7	11.5	10.3	7.0	5.6	(38)
(39)			MCWBR	4.9	6.6	7.0	6.8	7.2	6.6	6.8	6.2	6.7	6.8	5.7	4.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.315	0.337	0.375	0.411	0.413	0.409	0.428	0.420	0.384	0.358	0.335	0.309	(40)	
(41)		taud	2.375	2.355	2.260	2.207	2.239	2.271	2.229	2.265	2.355	2.410	2.406	2.372	(41)	
(42)		Ebn at Noon	684	775	808	822	840	846	820	804	793	739	652	623	(42)	
(43)		Edh at Noon	60	81	108	128	130	127	131	120	98	77	59	52	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.68	1.38	2.45	3.99	4.91	5.29	4.96	4.23	3.09	1.76	0.80	0.52	(44)	
(45)		RadStd	0.07	0.19	0.26	0.48	0.57	0.43	0.67	0.37	0.37	0.19	0.10	0.05	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (46 neighbors)		N/A	N/A	N/A	N/A	N/A	+0.40	N/A	N/A	N/A	N/A	

Nomenclature: See separate page